



Figure 1: Disk selection in PhotoRec



Figure 2: Partition selection

not impossible for experts). Thus, when you find that you have accidentally messed up and lost data that you want, you should avoid writing new data to the medium. This means you shouldn't continue running the OS that is using that partition or medium in read-write mode; even applications like a browser caching files, the OS downloading and applying updates in the background, or downloading new files and installing new packages, could cause the space of the deleted file to be overwritten. Therefore, shut down that OS immediately, and boot with a live CD or a live USB that does not write to that medium.



Tip: CGSecurity has a list of live CD images, which includes their utilities, at http://www.cgsecurity.org/wiki/TestDisk_Livecd. You could consider keeping a CD of one of these in your toolkit to guard against such emergencies.

Overview—TestDisk

This software should be used to recover lost partitions, or to make non-bootable disks bootable again. TestDisk claims to perform the following operations:

- Fix partition tables and recover deleted partitions
- Recover the FAT32 boot sector from its back-up
- Rebuild the FAT12/FAT16/FAT32 boot sector
- Fix FAT tables
- Rebuild the NTFS boot sector
- Recover the NTFS boot sector from its back-up
- Locate the ext2/ext3/ext4 back-up superblock
- Undelete files from the FAT, exFAT, NTFS and ext2 filesystem
- Copy files from the deleted FAT, exFAT, NTFS and ext2/ext3/ext4 partitions.

Overview—PhotoRec

This software claims to recover individual files, such as ZIP, Office, PDF, HTML, JPEG and 390 other file extensions. In case TestDisk fails to do the job, use PhotoRec. Personally, I prefer PhotoRec.



Figure 3: File type information



Figure 4: File type selection



Figure 5: Filesystem type selection

Using PhotoRec

PhotoRec recovers files by finding deleted files and copying them to another disk/medium (files should not be recovered to the partition/medium which has the deleted files because that could lead to data being overwritten, as explained before). Also, PhotoRec will most likely recover a lot of files, so the partition/medium on which you are storing recovered data should ideally be at least as large as the partition/medium being recovered.

The possible modes for recovery are:

- Recover the files to a separate hard drive.
 - Recover the files to a networked storage drive.
 - Recover the files to a separate partition on the same hard drive.
- I personally prefer the first option. Now, I have an 8 GB pen drive containing some data that I have formatted. I will show you how to recover data from it, using my laptop that runs Ubuntu 11.10.
1. First, download PhotoRec from the link mentioned earlier, and unpack it.
 2. Next, fire up the terminal, and navigate to the directory where it's extracted. Launch PhotoRec with: `sudo ./photorec_static`
 3. Now, select the disk from which to recover data. In my case, it is my HP pen drive, shown as `/dev/sdb` (Figure 1).
 4. Next, select a partition, as shown in Figure 2. You can also